Math 110 Hard Mock Midterm 2

Notes Ch. 8–13 50 minutes 4 questions 40 points

Set a 50-minute timer. Give concise reasoning with each answer. You may quote results from the notes or text if you state them. $\mathcal{P}_n(\mathbb{R})$ is the space of real polynomials of degree at most n.

1. Duality on polynomials (10 points) On $V = \mathcal{P}_2(\mathbb{R})$ define

$$\varphi_1(p) = p(1), \qquad \varphi_2(p) = p'(1), \qquad \varphi_3(p) = \int_0^2 p(x)\,dx.$$

a) (3) Find the basis p_1, p_2, p_3 of V whose dual basis is $\varphi_1, \varphi_2, \varphi_3$. (This proves that $\varphi_1, \varphi_2, \varphi_3$ is a basis of V'.)

b) (3) Let $U = \{p \in V : p(1) = 0 \text{ and } \int_0^2 p = 0\}$. Find a basis of U and a basis of U^0.

c) (4) Let $D \in \mathcal{L}(V)$ be $Dp = p'$. Write $D'\varphi_1$, $D'\varphi_2$, $D'\varphi_3$ as combinations of $\varphi_1, \varphi_2, \varphi_3$. Find $\operatorname{null} D'$ and check that it equals $(\operatorname{range} D)^0$.

2. Invariant complements (10 points) Let V be a nonzero finite-dimensional vector space and $T \in \mathcal{L}(V)$. Say T has *property (C)* if every T-invariant subspace U has a T-invariant W with $V = U \oplus W$.

a) (3) Prove: if T is diagonalizable, then T has property (C).

b) (3) Let V be complex and suppose T has property (C). Let U be the span of all eigenvectors of T. Prove $U = V$, and conclude that T is diagonalizable.

c) (2) Show that (b) fails over $\mathbb{R}$.

d) (2) For $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ on $\mathbb{C}^2$, find an invariant subspace with no invariant complement.

3. Shifts and quotients (10 points) On $V = \mathcal{P}_3(\mathbb{R})$ let D be differentiation, $(Ep)(x) = p(x+1)$, and $U = \mathcal{P}_1(\mathbb{R})$.

a) (3) Prove that U is invariant under D and under E. Find the matrices of D/U and E/U in the basis $x^2 + U,\ x^3 + U$ of V/U.

b) (3) Find $m_{D/U}$ and $m_{E/U}$. Are D/U and E/U diagonalizable?

c) (4) Prove $E = I + D + \frac{1}{2}D^2 + \frac{1}{6}D^3$, and deduce that $m_E = (x-1)^4$.

4. Minimal polynomials under duality and quotients (10 points) Let V be finite-dimensional and $T \in \mathcal{L}(V)$.

a) (3) Prove that $p(T)' = p(T')$ for every polynomial p, and that $S' = 0$ iff $S = 0$ for $S \in \mathcal{L}(V)$. Deduce $m_{T'} = m_T$.

b) (3) Let U be T-invariant. Prove that U^0 is T'-invariant and that the minimal polynomial of $T'|_{U^0}$ divides m_T.

c) (4) Now let V be complex with $\dim V = 3$, $m_T = (x-1)^2(x-2)$, and $U = E(2, T)$. Prove $\dim U = 1$ and find $m_{T/U}$.